Data Structure (C Programming)

25/08/2026

Name : Harshad Sanjay Teli.
Roll No : 15
Batch : B1
Prerequisite
Based on Array.

Q1. Write a C program to find the largest and smallest element in an array of integers.
Input:

Que 1] Based on Array.
Write a C program to find the largest and smallest element in an array of integers.

→

```
#include<stdio.h>
int main()
{
  int a[100],size,i,max;
  printf(" Enter Size of Array: ");
  scanf("%d", &size);
  for(i=0; i<size; i++)
  {
     scanf(" %d", &a[i]);
  }
   max=a[0];
   for(i=0; i<size; i++)
```

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```
{ if a[i]>max)
 { max=a[i];
}}
printf(" Maximum Value: %d",
//similarly  min=a[0]
for(i=0; i<size; i++)
 { if(a[i]<min)
 {  min=a[i];
}}
Printf(" Minimum element: %d", min);
 return 0;
}.
```

45   7

Output:

```
PS C:\Users\Harshad Teli\Downloads\DS> gcc -o output1 Program.c
PS C:\Users\Harshad Teli\Downloads\DS> ./output1
Enter Size of Array:5
7
6
3
8
1
Minimum Value from the Array is 1
Maximum Value from the Array is 8
PS C:\Users\Harshad Teli\Downloads\DS>
```

Q2. Write a C program to reverse the elements of an array.

Input:

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Q.2] Write a C Program to reverse the elements of an Array

```
#include<stdio.h>
int main()
{
   int arr[10];
   printf("Enter Array Elements:\n");
   for(int i=0; i<10; i++)
   {
      scanf("%d", &arr[i]);
   }
   printf("Reverse of the Array Element is\n");
   for(int i=9; i>=0; i--)
   {
      printf("%d\n", arr[i]);
   }
   return 0;
}
```

O/P:- Enter Array Elements with size of 5:

```
2
6
78
56
89
```

The reverse elements of the Array is:

```
89
56
78
6
2
```

Output:

```
student@student-OptiPlex-SFF-7020:~/Pictures/MCA_15$ gcc -o output Classroom.c
student@student-OptiPlex-SFF-7020:~/Pictures/MCA_15$ ./output
Enter Array Elements with the Size of 5:
2
6
78
56
89
The reverse elements of the Array is :
89
56
78
6
2
student@student-OptiPlex-SFF-7020:~/Pictures/MCA_15$
```

Q3. Write a C program to perform the addition of two matrices (2D arrays) of size 3×3.

Input:

Q3 Write a C Program to Perform the addition of two matrices (2D) of sizes 3X3.

→

```
#include<stdio.h>
int main()
{
  int arr1[3][3];
  int arr2 [3][3];
  int result(3][3];
  printf("Enter elements of first Array:");
  for(int i= 0;i<=2;i++)
  {
    for (int j=0;j<=2;j++)
    {
      scanf("%d",&arr1[i][j]);
    }
  }
  printf("Enter elements of Second Array:");
  for(int i=0;i<=2;i++)
  {
    for(int j=0;j<=2;j++)
    {
      scanf("%d",&arr2(i][j]);
    }
  }.
  printf("The Addition of the Array is :\n");
  for(int i=0;i<2;i++)
  { for(int j=0;j<2;j++)
    {
      result[i][j]= arr1[i][j]+ arr2[i][j];
      printf("%d\t",result[i][j]);
    }
    printf("\n");
  }
  return 0;
}.
```

Name: Harshad Teli Sign:

Roll No:15

Output:

```
Enter the elements of the Array1 with the Size 3X3:

Enter the element for the arr[0][0]=6

Enter the element for the arr[0][1]=23

Enter the element for the arr[0][2]=78

Enter the element for the arr[1][0]=96

Enter the element for the arr[1][1]=2

Enter the element for the arr[1][2]=5

Enter the element for the arr[2][0]=15

Enter the element for the arr[2][1]=36

Enter the element for the arr[2][2]=45
Enter the elements of the Array2 with the Size 3X3:

Enter the element for the arr[0][0]=10

Enter the element for the arr[0][1]=2

Enter the element for the arr[0][2]=36

Enter the element for the arr[1][0]=48

Enter the element for the arr[1][1]=2

Enter the element for the arr[1][2]=9

Enter the element for the arr[2][0]=3

Enter the element for the arr[2][1]=47

Enter the element for the arr[2][2]=2
The Addition of the Array is:
16      25      114
144     4       14
18      83      47
```